11th grade
Learning evidence 2.1: Confidence Interval Planning (Solutions)

Problems

Problem 1:
Problem description

A city budgeting office wants to estimate the average weekly transit subsidy per rider for two pilot zones. The two samples are different random samples drawn from the same underlying population of weekly subsidy amounts. Zone A recorded the following sample of weekly subsidy amounts (USD):

- Zone A sample ($n_A = 6$): 18, 22, 19, 21, 20, 17.
- Zone B sample ($n_B = 3$): 24, 26, 23.

An annual audit provides the population standard deviation for this population: $\sigma = 3{,}5$. For classroom purposes, suppose the true population mean weekly subsidy for this population is μ. Construct and interpret a 95 % confidence interval for the population mean weekly subsidy in each zone.

C1

First, we calculate the sample means.

$$\bar{x}_A = \frac{\sum x_A}{n_A}$$

$$\sum x_A = 18 + 22 + 19 + 21 + 20 + 17 = 117$$

$$\bar{x}_A = \frac{117}{6} = 19{,}5$$

$$\bar{x}_B = \frac{\sum x_B}{n_B}$$

$$\sum x_B = 24 + 26 + 23 = 73$$

$$\bar{x}_B = \frac{73}{3} \approx 24{,}33$$

Second, we calculate the sample standard deviations.

$$s_A^2 = \frac{\sum (x_A - \bar{x}_A)^2}{n_A - 1}$$

$$\sum (x_A - \bar{x}_A)^2 = (18 - 19{,}5)^2 + (22 - 19{,}5)^2 + (19 - 19{,}5)^2 + (21 - 19{,}5)^2 + (20 - 19{,}5)^2 + (17 - 19{,}5)^2 = 17{,}5$$

$$s_A = \sqrt{\frac{17{,}5}{6 - 1}} = \sqrt{3{,}5} \approx 1{,}87$$

$$s_B^2 = \frac{\sum (x_B - \bar{x}_B)^2}{n_B - 1}$$

$$\sum (x_B - \bar{x}_B)^2 = (24 - 24{,}33)^2 + (26 - 24{,}33)^2 + (23 - 24{,}33)^2 \approx 4{,}67$$

$$s_B = \sqrt{\frac{4{,}67}{3 - 1}} \approx 1{,}53$$

C2

The sample statistics are $\bar{x}_A = 19{,}5$, $s_A \approx 1{,}87$, $\bar{x}_B \approx 24{,}33$, and $s_B \approx 1{,}53$, while the population parameters are μ and $\sigma = 3{,}5$. Both samples come from the same population, so each sample provides an estimate for μ.

C3

A single point estimate can miss the true mean because of sampling variation. A confidence interval shows a range of plausible values for the mean μ, which is more informative for planning.

C5

With known $\sigma = 3{,}5$ and $n_A = 6$,

$$SE_A = \frac{\sigma}{\sqrt{n_A}}$$

$$SE_A = \frac{3{,}5}{\sqrt{6}} \approx 1{,}43$$

For 95 % confidence, $z_{0{,}025} = 1{,}96$.

$$E_A = z_{0{,}025} \cdot SE_A$$

$$E_A = 1{,}96(1{,}43) \approx 2{,}80$$

$$\mu \in \bar{x}_A \pm E_A$$

$$\mu \in 19{,}5 \pm 2{,}80 = [19{,}5 - 2{,}80, 19{,}5 + 2{,}80] = [16{,}70, 22{,}30]$$

With known $\sigma = 3{,}5$ and $n_B = 3$,

$$SE_B = \frac{\sigma}{\sqrt{n_B}}$$

$$SE_B = \frac{3{,}5}{\sqrt{3}} \approx 2{,}02$$

$$E_B = z_{0{,}025} \cdot SE_B$$

$$E_B = 1{,}96(2{,}02) \approx 3{,}96$$

$$\mu \in \bar{x}_B \pm E_B$$

$$\mu \in 24{,}33 \pm 3{,}96 = [24{,}33 - 3{,}96, 24{,}33 + 3{,}96] = [20{,}37, 28{,}29]$$

C4

We are 95 % confident the true mean weekly subsidy in the population is between about 16.70 and 22.30 dollars based on Sample A, and 95 % confident it is between about 20.37 and 28.29 dollars based on Sample B. The intervals reflect uncertainty from different samples drawn from the same population.

Problem 2: ## Problem description

A regional warehouse studies the number of orders processed per hour to plan staffing. A random sample of 54 hours is summarized in grouped form below. The operations team also reports a known population standard deviation of $\sigma = 6{,}2$ orders per hour. For academic purposes only, assume the true population mean order rate is $\mu = 22$ orders per hour. The grouped data pairs list orders per hour first and the frequency (number of hours) second: $14 \to 8$, $18 \to 16$, $22 \to 20$, and $26 \to 10$. Construct and interpret a 90 % confidence interval and a 98 % confidence interval for the population mean orders per hour.

C1

Using grouped values x with frequencies f.

$$\bar{x} = \frac{\sum fx}{n}$$

$$n = \sum f$$

$$n = 8 + 16 + 20 + 10 = 54$$

$$\sum fx = 8(14) + 16(18) + 20(22) + 10(26) = 1100$$

$$\bar{x} = \frac{1100}{54} \approx 20{,}37$$

Second, we calculate the sample standard deviation s.

$$s^2 = \frac{\sum f(x - \bar{x})^2}{n - 1}$$

$$\sum f(x - \bar{x})^2 \approx 8(14 - 20{,}37)^2 + 16(18 - 20{,}37)^2 + 20(22 - 20{,}37)^2 + 10(26 - 20{,}37)^2$$

$$\sum f(x - \bar{x})^2 \approx 324{,}65 + 89{,}90 + 53{,}11 + 316{,}93 = 784{,}59$$

$$s = \sqrt{\frac{\sum f(x - \bar{x})^2}{n - 1}}$$

$$s = \sqrt{\frac{784{,}59}{54 - 1}} \approx 3{,}85$$

C2

The sample statistics are $\bar{x} \approx 20{,}37$ and $s \approx 3{,}85$, while the population parameters are μ and $\sigma = 6{,}2$. The sample values estimate the population values.

C3

The sample summarizes only 54 hours, so the true mean could differ from the point estimate. A confidence interval expresses the likely range for μ based on the data.

C5

With known $\sigma = 6{,}2$ and $n = 54$,

$$SE = \frac{\sigma}{\sqrt{n}}$$

$$SE = \frac{6{,}2}{\sqrt{54}} \approx 0{,}84$$

For 90 % confidence, $z_{0{,}05} = 1{,}645$.

$$E_{90} = z_{0{,}05} \cdot SE$$

$$E_{90} = 1{,}645(0{,}84) \approx 1{,}39$$

$$\mu \in \bar{x} \pm E_{90}$$

$$\mu \in 20{,}37 \pm 1{,}39 = [20{,}37 - 1{,}39, 20{,}37 + 1{,}39] = [18{,}98, 21{,}76]$$

For 98 % confidence, $z_{0{,}01} = 2{,}326$.

$$E_{98} = z_{0{,}01} \cdot SE$$

$$E_{98} = 2{,}326(0{,}84) \approx 1{,}96$$

$$\mu \in \bar{x} \pm E_{98}$$

$$\mu \in 20{,}37 \pm 1{,}96 = [20{,}37 - 1{,}96, 20{,}37 + 1{,}96] = [18{,}41, 22{,}33]$$

C4

We are 90 % confident the true mean orders processed per hour is between about 18.98 and 21.76 orders, and 98 % confident it is between about 18.41 and 22.33 orders. The intervals show the range supported by the sample and known standard deviation.

Problem 3: Problem description

A logistics firm monitors the time (in minutes) required to load containers at a port. A sample of 55 loading times is grouped into class intervals below. The firm has a historical estimate of the population standard deviation, $\sigma = 5{,}8$ minutes. The grouped intervals and frequencies are 25–29 minutes (18), 30–34 minutes (20), and 35–39 minutes (17). Construct and interpret a 95 % confidence interval for the population mean loading time.

C1

Using grouped midpoints x with frequencies f.

$$\bar{x} = \frac{\sum fx}{n}$$

$$n = \sum f$$

$$n = 18 + 20 + 17 = 55$$

$$\sum fx = 18(27) + 20(32) + 17(37) = 1755$$

$$\bar{x} = \frac{1755}{55} \approx 31{,}91$$

Second, we calculate the sample standard deviation s.

$$s^2 = \frac{\sum f(x-\bar{x})^2}{n-1}$$

$$\sum f(x-\bar{x})^2 \approx 18(27-31{,}91)^2 + 20(32-31{,}91)^2 + 17(37-31{,}91)^2$$

$$\sum f(x-\bar{x})^2 \approx 874{,}55$$

$$s = \sqrt{\frac{\sum f(x-\bar{x})^2}{n-1}}$$

$$s = \sqrt{\frac{874{,}55}{55-1}} \approx 4{,}02$$

C2

The sample statistics are $\bar{x} \approx 31{,}91$ and $s \approx 4{,}02$, while the population parameters are μ and $\sigma = 5{,}8$. The sample values estimate the population values, and μ remains unknown.

C3

Even with 55 observations, a point estimate does not show how much the mean could vary. A confidence interval summarizes that uncertainty by giving a plausible range for μ.

C5

With known $\sigma = 5{,}8$ and $n = 55$,

$$SE = \frac{\sigma}{\sqrt{n}}$$

$$SE = \frac{5{,}8}{\sqrt{55}} \approx 0{,}78$$

For 95 % confidence, $z_{0{,}025} = 1{,}96$.

$$E = z_{0{,}025} \cdot SE$$

$$E = 1{,}96(0{,}78) \approx 1{,}53$$

$$\mu \in \bar{x} \pm E$$

$$\mu \in 31{,}91 \pm 1{,}53 = [31{,}91 - 1{,}53, 31{,}91 + 1{,}53] = [30{,}38, 33{,}44]$$

C4

We are 95 % confident the true mean loading time is between about 30.38 and 33.44 minutes. The interval provides a plausible range for the population mean based on the sample.